

EXPERIMENT – 1 Interfacing With Arduino

PROJECT NAME

Weather Forecasting System with Extreme Temperature Indicator

OBJECTIVE / PROBLEM STATEMENT

This project monitors environmental temperature and humidity using a DHT11 sensor and predicts weather conditions based on temperature values.

The system uses RGB LED indicators to show:

- Extreme Hot Weather
- Normal Weather
- Cold Weather

A DS1307 RTC module provides real-time clock information, and an SSD1306 OLED display shows:

- Current Time
- Temperature
- Humidity
- Weather Status

A push button is also used as a master power control for turning the complete system ON and OFF.

COMPONENTS USED

Component	Quantity
Arduino UNO	1
DHT11 Sensor	1
DS1307 RTC Module	1
SSD1306 OLED Display	1
RGB LED	1
Push Button	1
Resistors (100Ω)	3
Jumper Wires	Required
Breadboard	1

PIN CONNECTIONS

Device	Arduino Pin
DHT11 Data Pin	D7
Red LED	D2
Green LED	D3
Blue LED	D5
Power Button	D8
OLED SDA	A4
OLED SCL	A5
RTC SDA	A4
RTC SCL	A5

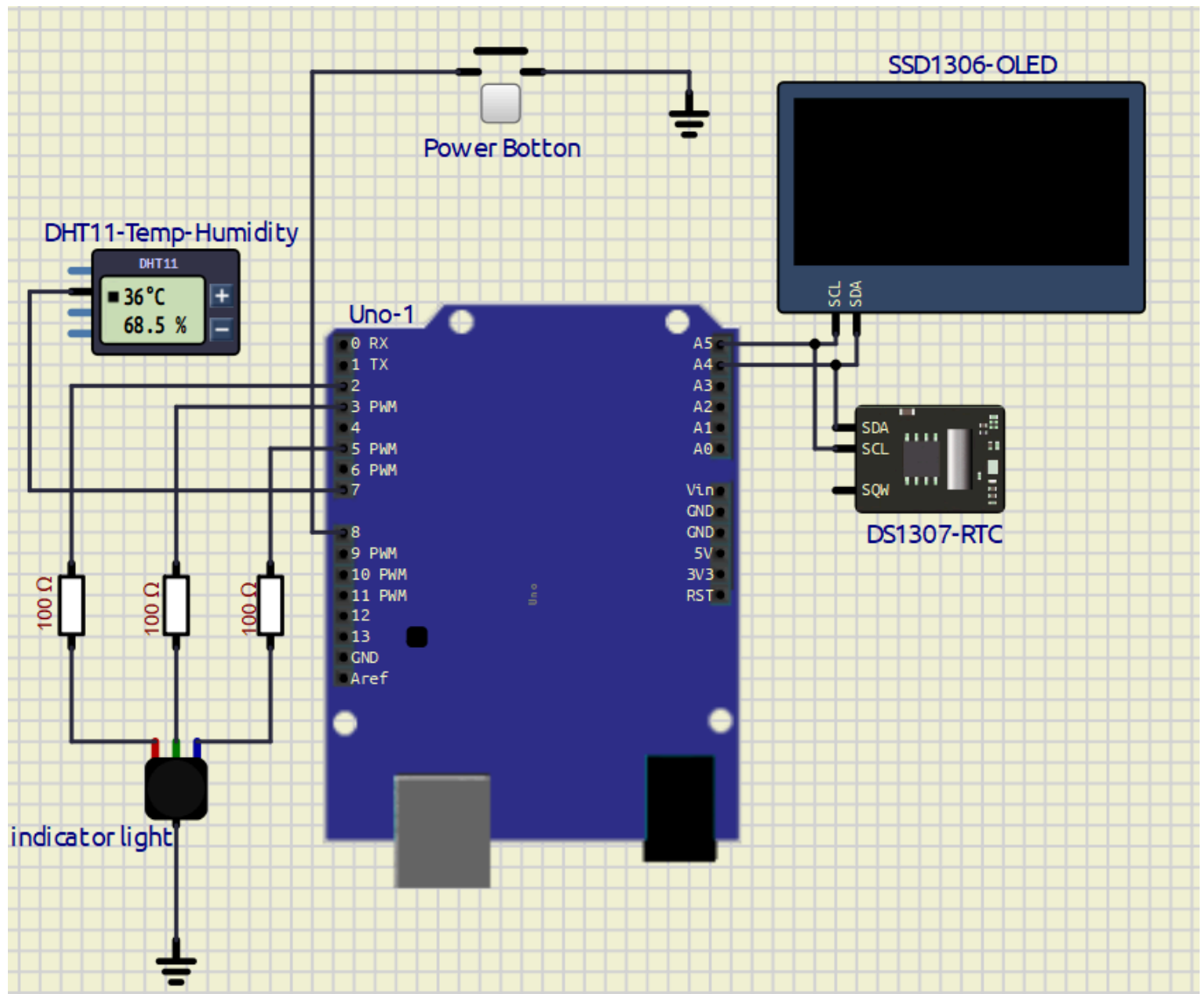
RGB LED CONNECTION

RGB Pin	Connection
Red Pin	D2 through 100Ω
Green Pin	D3 through 100Ω
Blue Pin	D5 through 100Ω
Common Cathode	GND

SOFTWARE USED

- Arduino IDE
- SimulIDE

CIRCUIT DIAGRAM



REQUIRED LIBRARIES

```
#include <Wire.h>
#include <RTClib.h>
#include <DHT.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
```

MAIN ARDUINO CODE

Code

```
#include <Wire.h>
#include <RTClib.h>

#include <DHT.h>
```

```
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64

Adafruit_SSD1306 display(
  SCREEN_WIDTH,
  SCREEN_HEIGHT,
  &Wire,
  -1
);

RTC_DS1307 rtc;

#define DHTPIN 7
#define DHTTYPE DHT11

DHT dht(DHTPIN, DHTTYPE);

#define RED_LED      2
#define GREEN_LED    3
#define BLUE_LED     5

#define POWER_BUTTON 8

bool systemState = false;

bool lastButtonState = HIGH;

void setup()
{
  pinMode(RED_LED, OUTPUT);

  pinMode(GREEN_LED, OUTPUT);

  pinMode(BLUE_LED, OUTPUT);

  pinMode(POWER_BUTTON, INPUT_PULLUP);

  Wire.begin();

  rtc.begin();

  dht.begin();

  // rtc.adjust(DateTime(F(__DATE__), F(__TIME__)));

  if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C))
  {
    while (1);
  }

  display.clearDisplay();
```

```
    display.display();
}

void loop()
{
    bool buttonState = digitalRead(POWER_BUTTON);

    if (
        buttonState == LOW &&
        lastButtonState == HIGH
    )
    {
        delay(50);

        systemState = !systemState;

        if (systemState == true)
        {
            display.clearDisplay();

            display.setTextSize(2);

            display.setTextColor(WHITE);

            display.setCursor(10, 25);

            display.println("POWER ON");

            display.display();

            delay(1000);
        }

        else
        {
            display.clearDisplay();

            display.setTextSize(2);

            display.setTextColor(WHITE);

            display.setCursor(10, 25);

            display.println("POWER OFF");

            display.display();

            delay(1000);

            display.clearDisplay();

            display.display();
        }
    }
}
```

```
}

lastButtonState = buttonState;

if (systemState == false)
{
    digitalWrite(RED_LED, LOW);

    digitalWrite(GREEN_LED, LOW);

    digitalWrite(BLUE_LED, LOW);

    return;
}

DateTime now = rtc.now();

float temperature = dht.readTemperature();

float humidity = dht.readHumidity();

String statusText;

if (temperature > 35)
{
    digitalWrite(RED_LED, HIGH);

    digitalWrite(GREEN_LED, LOW);

    digitalWrite(BLUE_LED, LOW);

    statusText = "EXTREME HOT";
}

else if (temperature < 20)
{
    digitalWrite(RED_LED, LOW);

    digitalWrite(GREEN_LED, LOW);

    digitalWrite(BLUE_LED, HIGH);

    statusText = "COLD";
}

else
{
    digitalWrite(RED_LED, LOW);

    digitalWrite(GREEN_LED, HIGH);

    digitalWrite(BLUE_LED, LOW);

    statusText = "NORMAL";
}
```

```
}

display.clearDisplay();

display.setTextSize(1);

display.setTextColor(WHITE);

display.setCursor(0, 0);

display.println("WEATHER SYSTEM");

display.setCursor(0, 15);

display.print("TIME : ");

print2digit(now.hour());

display.print(":");

print2digit(now.minute());

display.print(":");

print2digit(now.second());

display.setCursor(0, 30);

display.print("TEMP : ");

display.print(temperature);

display.println(" C");

display.setCursor(0, 42);

display.print("HUM : ");

display.print(humidity);

display.println(" %");

display.setCursor(0, 55);

display.print("STATUS:");

display.println(statusText);

display.display();

delay(50);
}
```

```
void print2digit(int number)
```

```
{  
  if (number < 10)  
  {  
    display.print("0");  
  }  
  
  display.print(number);  
}
```

WORKING PRINCIPLE

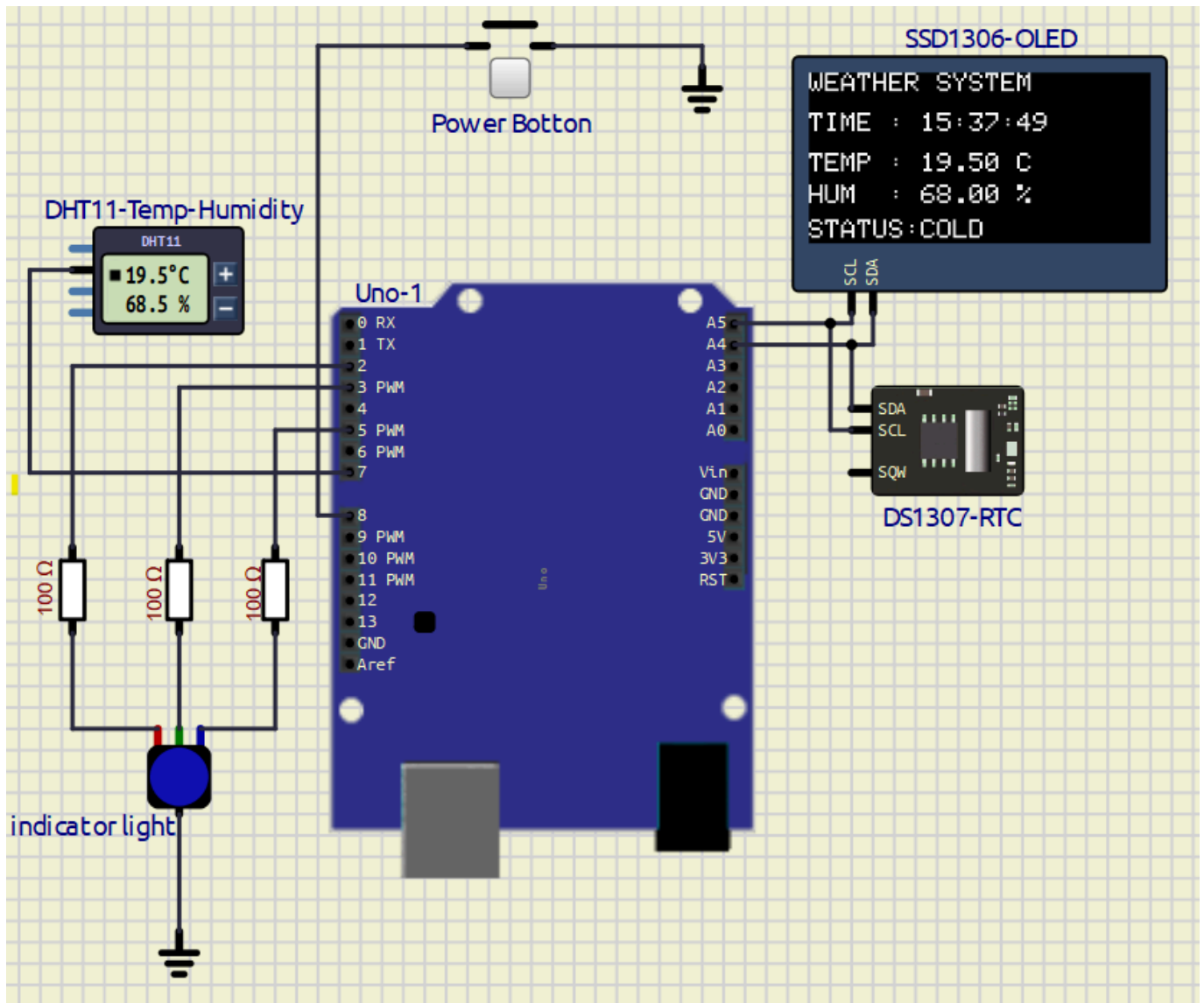
1. The system starts with a blank OLED screen.
2. Pressing the power button:
 - Turns the system ON
 - Displays "POWER ON"
3. DHT11 sensor continuously reads:
 - Temperature
 - Humidity
4. DS1307 RTC module provides:
 - Real-time clock
5. OLED displays:
 - Time
 - Temperature
 - Humidity
 - Weather Status
6. RGB LED indicates weather condition:
 - Red LED → Extreme Hot
 - Green LED → Normal Weather
 - Blue LED → Cold Weather
7. Pressing the power button again:
 - Displays "POWER OFF"
 - Turns OFF all LEDs
 - OLED returns to black screen

TEMPERATURE INDICATOR LOGIC

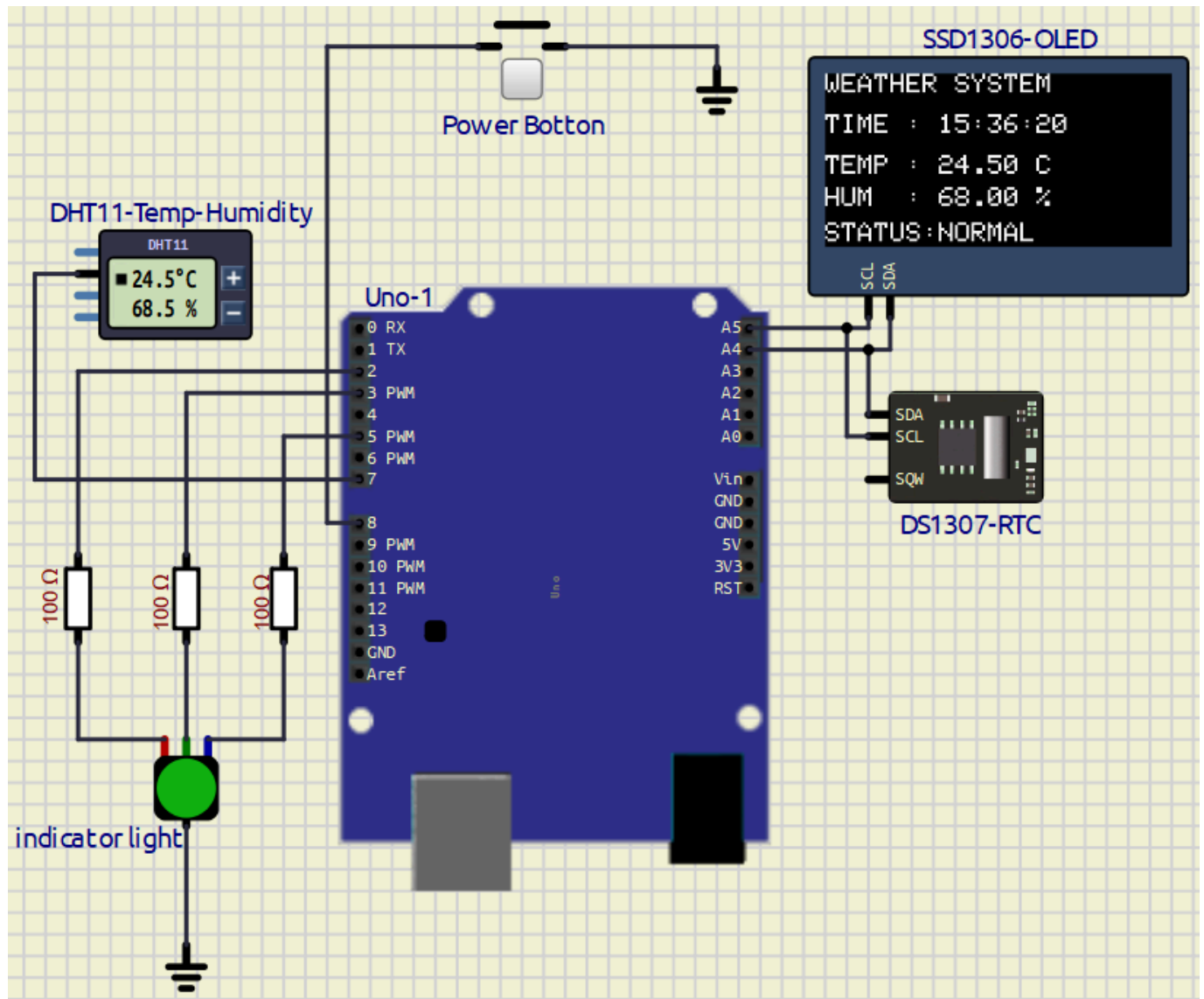
Temperature	LED Status	Weather
> 35°C	Red LED ON	Extreme Hot
20°C – 35°C	Green LED ON	Normal
< 20°C	Blue LED ON	Cold

OUTPUT

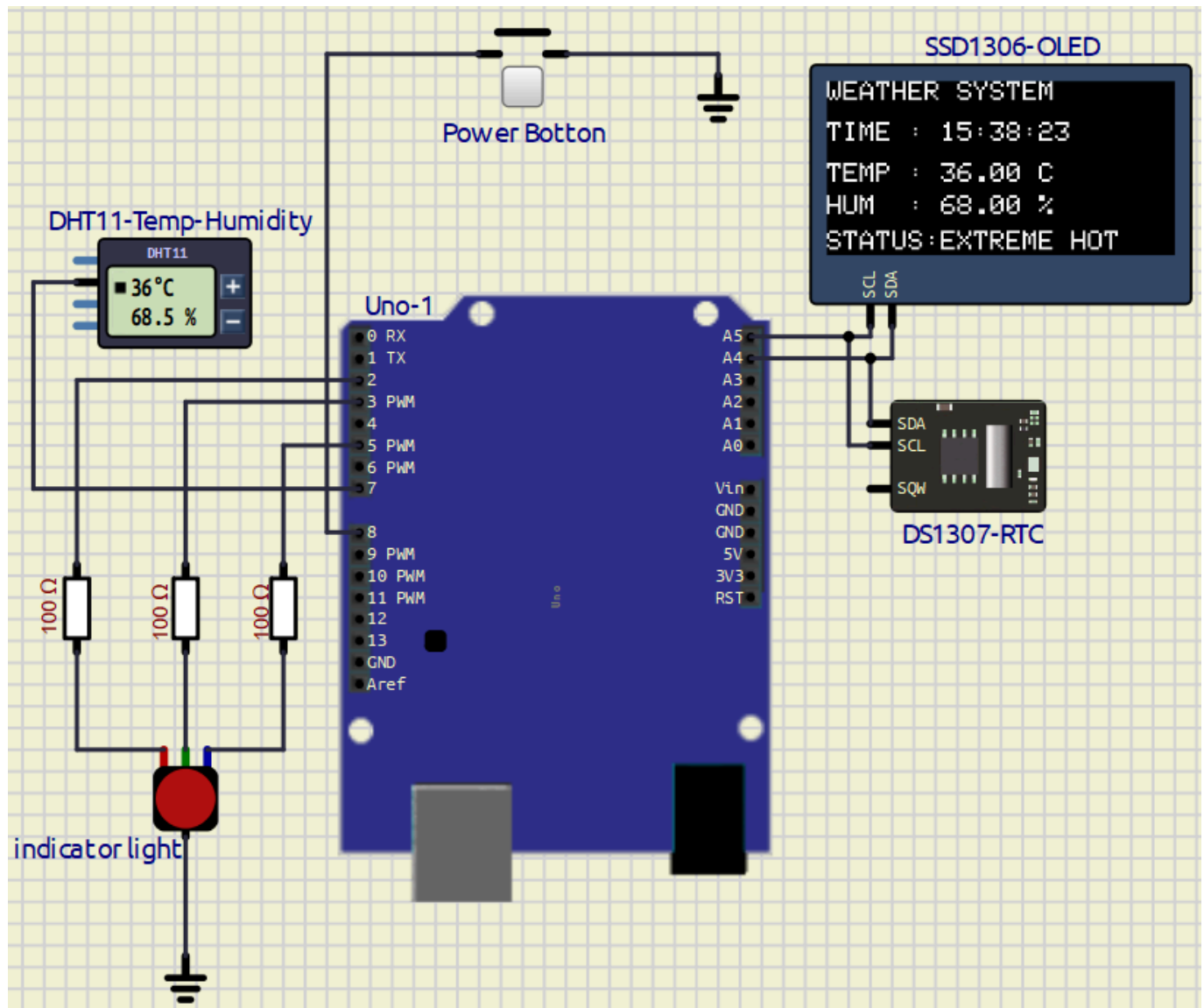
System - COLD (Temp < 20)



System - Normal (Temp > 20 & < 36)



System - Extream Hot (Temp > 35)



ADVANTAGES

- Real-time weather monitoring
- Simple embedded system design
- RGB visual indication
- RTC-based real-time display
- Easy ON/OFF control
- Low-cost implementation
- User-friendly interface

APPLICATIONS

- Smart weather stations
- Environmental monitoring
- Temperature warning systems
- Industrial monitoring
- Home automation systems
- Embedded systems learning

CONCLUSION

The Weather Forecasting System using Arduino UNO, DHT11 sensor, DS1307 RTC module, SSD1306 OLED display, RGB LED, and push-button control was successfully implemented in SimulIDE.

The system correctly monitors weather conditions, displays real-time environmental information, and visually indicates temperature conditions using RGB LEDs.

During this experiment, I learned:

- Sensor interfacing
- OLED interfacing
- RTC interfacing
- RGB LED control
- Push-button logic
- Embedded system design

REFERENCES

1. Arduino UNO Documentation
<https://docs.arduino.cc/hardware/uno-rev3/>
2. DHT11 Sensor Documentation
<https://components101.com/sensors/dht11-temperature-sensor>
3. DS1307 RTC Datasheet
<https://datasheets.maximintegrated.com/en/ds/DS1307.pdf>
4. Adafruit SSD1306 Library
https://github.com/adafruit/Adafruit_SSD1306
5. Adafruit GFX Library
<https://github.com/adafruit/Adafruit-GFX-Library>
6. RTCLib Documentation
<https://github.com/adafruit/RTCLib>

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